The Journal of Open Source Software

RMAVIS v1.0: a R Shiny application for the analysis of vegetation survey data and assignment to GB NVC communities.

Zeke Marshall [1], **Simon Mark Smart** [1], **Colin Harrower** [2], **and Rob Marrs** [3]

1 UK Centre for Ecology & Hydrology, Lancaster Environment Centre, Library Avenue, Bailrigg, Lancaster, LA1 4AP, United Kingdom **2** UK Centre for Ecology & Hydrology, Maclean Building, Benson Lane, Crowmarsh Gifford, Wallingford, Oxfordshire, OX10 8BB, United Kingdom **3** School of Environmental Sciences, University of Liverpool, Liverpool, L69 3GP, UK

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Michael Mahoney

Reviewers:

- @adithirgis
- @rasanderson

Submitted: 12 April 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

RMAVIS is a R Shiny application for the assignment of vegetation sample plot data to British National Vegetation Classification (NVC) communities (Rodwell, 1991, 1992a, 1992b, 1995, 2000).

The assignment of vegetation sample plot data to established vegetation classification units using computational methods is a well established and recognised practice (Maciejewski et al., 2020). The results of this assignment process are used in various ways, including assisting in the phase 2 habitat survey (or NVC survey) (Rodwell, 2006) process; establishing an ecological baseline and identifying important ecological features such as protected habitats (CIEEM, 2022); and in ecological restoration by providing a proxy for historical reference ecosystems to target and against which to measure restoration progress (Gann et al., 2019; Pywell et al., 2002; Sturbois et al., 2023).

In Great Britain (GB) the development of computational methods and programs for the assignment of vegetation survey data to NVC communities began with the development of TABLEFIT (Hill, 1989; Marrs, 2019) and was followed by MATCH (Malloch, 1998). The most recent program, the Modular Analysis of Vegetation Information System (MAVIS), was developed as a windows application in 2000 and was a DEFRA-funded output of the ECOFACT project (Bunce et al., 1999), with the latest version released in 2016 (Smart et al., 2016).

Statement of need

The requirement for a new program in the form of a R Shiny (Chang et al., 2024) application arises from several needs, namely to: 1) accommodate updates to the NVC; 2) provide a means to easily reproduce and attribute the results of the NVC assignment process; 3) broaden and simplify access through the provision of a web-based Graphical User Interface (GUI); and 4) facilitate the continuous development of such NVC assignment software. We developed RMAVIS considering these needs, with the view to providing a reliable system for use by the GB ecology and conservation community, analogous to the Engine for Relevés to Irish Communities Assignment (ERICA) tool (Perrin et al., 2018; Perrin, 2019) developed for Ireland, and the EuroVegChecklist Expert System (EVC-ES) (Mucina et al., 2016) developed for Europe.

The Journal of Open Source Software

Application structure

Inspired by the extensible structure of the species niche and distribution modelling R shiny application `wallace` (Kass et al., 2023), we constructed `RMAVIS` with a modular architecture, facilitating both the maintenance of existing modules and development of new modules with ease.

`RMAVIS` currently contains a total of twenty-one modules, with the fourteen main modules summarised in the following table.

Module	Description
Sidebar	Acts as the control module, containing the options for each module.
Data Input	Facilitates the entry of data, three methods are provided: manual, upload, and example.
Data Validation	Checks the format of the inputted data and provides means to adjust incorrectly formatted entries.
Data Structure	Checks the structure of the inputted data, displaying the availability of data by species and number of plots per year and group.
NVC Assignment	Displays the results of the NVC assignment process.
Habitat Correspondence	Displays the habitats from alternative habitat classifications associated with the top-fitted NVC communities.
Floristic Tables	Allows the comparison of floristic tables composed from the inputted data with the NVC's floristic tables.
Frequency	Summarises the frequency of occurrence of each species across all plots over time.
EIVs	Displays the mean Hill-Ellenberg ecological indicator values (EIVs) for each plot, group of plots, and all plots.
Diversity	Displays a number of diversity metrics for each plot and group of plots.
MVA National	Constructs an ordination space using all NVC pseudo-quadrats, with sample plots added in passively.
MVA Local, restricted	Constructs an ordination space using the top-fitted NVC communities pseudo-quadrats, with sample plots added in passively.
MVA Local, unrestricted	Constructs an ordination space using the top-fitted NVC communities pseudo-quadrats and sample plots.
Report	Provides the user with a downloadable report, containing user-selected outputs from the app session.

DRAFT

The Journal of Open Source Software

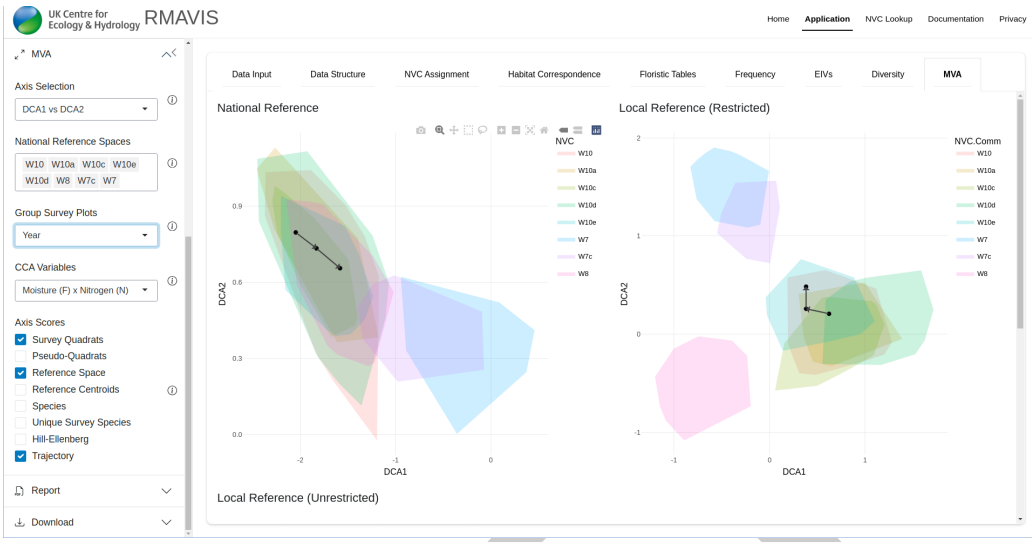

Figure 1: The MVA module of `RMAVIS`, showing the trajectory of all sample plots from the Leith Hill Wood example dataset in the ordination spaces over time.

Data sources and dependencies

`RMAVIS` depends on a number of individual datasets. The list of accepted species was constructed using the vascular plant (*Tracheophyta* and *Pteridophyta*), moss (*Bryophyta*), liverwort (*Marchantiophyta*), and hornwort (*Anthocerotophyta*) taxa present in the UKSI (Raper, 2015), retrieved from the National Biodiversity Network (NBN), (NBN Trust, 2024d, 2024e, 2024b, 2024c, 2024a) filtered to include taxa at the species, species hybrid, species aggregate, subspecies, species sensu lato, and genus ranks; along with the limited number of lichen (*Lecanoromycetes*), charophyte (*Charophyta*), and alga (*Algae*) taxa present in the NVC floristic tables. The NVC communities present in `RMAVIS` are composed from Rodwell (1991), Rodwell (1992a), Rodwell (1992b), Rodwell (1995), Rodwell (2000), Wallace & Prosser (2017), Prosser et al. (2023), and Wallace et al. (2023). Data for habitat correspondences is derived from: the Joint Nature Conservation Committee (JNCC) Spreadsheet of Habitat Correspondences (JNCC, 2008), UKHab V1.1 (Butcher et al., 2020), and the National Plant Monitoring Scheme (NPMS) habitat correspondences (Pescott et al., 2019). Data for the vascular plant Hill-Ellenberg values were retrieved from the BSBI checklists, specifically the Nitrogen Score (N) (BSBI, 2017c), Moisture Score (F) (BSBI, 2017d), Reaction Score (R) (BSBI, 2017a), Salinity Score (S) (BSBI, 2017b), and Light Score (L) (BSBI, 2017e) checklists. The corresponding data for bryophytes was taken from BRYOATT (Hill et al., 2007). Four example datasets are bundled with `RMAVIS`: 1) Parsonage Down (Ridding et al., 2020), 2) Leith Hill Place Wood (Smart et al., 2024; Wood et al., 2015), 3) Whitwell Common (Smart, 2000), and 4) Newborough Warren (Wallace & Jones, 2018).

Version 1.0 of `RMAVIS` was built under R 4.4.0 (R Core Team, 2024) and depends on the `bookdown` (Xie, 2024), `bsicons` (Sievert et al., 2023), `bslib` (Sievert et al., 2024), `dplyr` (Wickham et al., 2019), `ggplot2` (Wickham et al., 2019), `htmltools` (Cheng et al., 2024), `htmlwidgets` (Vaidyanathan et al., 2023), `kableExtra` (Zhu, 2024), `knitr` (Xie, 2024), `magrittr` (Stefan et al., 2022), `markdown` (Xie, 2023), `plotly` (Sievert, 2020), `purrr` (Wickham et al., 2019), `reactable` (Lin, 2024), `readr` (Wickham et al., 2019), `rhandsontable` (Owen, 2021), `rmarkdown` (Allaire et al., 2023), `shiny` (Chang et al., 2024), `shinybusy` (Meyer & Perrier, 2024), `shinyjs` (Attali, 2023), `shinyWidgets` (Perrier et al., 2024), `stringr`(Wickham et al., 2019), `tibble` (Wickham et al., 2019), `tidyr` (Wickham et al., 2019), `vegan` (Oksanen et al., 2022), and `writexl` (Ooms, 2024) R packages.

DRAFT

Conclusion

RMAVIS provides a web-based, easily accessible GUI for the assignment of vegetation sample plot data to GB NVC communities. RMAVIS also provides a number of other exploratory analyses, which compliment the NVC assignment results. We plan to maintain, optimise, and expand the functionality found in v1.0 of RMAVIS and hope that it acts as a useful tool for the GB ecology and conservation community.

Acknowledgements

The development of RMAVIS was partly supported by the UK-SCAPE program delivering National Capability (NE/R016429/1) funded by the Natural Environment Research Council.

We would like to thank Lindsay Maskell, Lucy Ridding, Barry Jobson, Colin Conroy, Rob Marrs, Andy McMullen, John Handley, Michael Tso, Simon Rolph, Cristina Martin Hernandez, and George Linney for testing RMAVIS.

We would also like to thank Laurence Jones for providing the Newborough Warren example dataset and Lucy Ridding for providing the Parsonage Down example dataset.

References

Allaire, J., Xie, Y., Dervieux, C., McPherson, J., Luraschi, J., Ushey, K., Atkins, A., Wickham, H., Cheng, J., Chang, W., & Iannone, R. (2023). *Rmarkdown: Dynamic Documents for R*. https://github.com/rstudio/rmarkdown

Attali, D. (2023). *Shinyjs: Easily Improve the User Experience of Your Shiny Apps in Seconds*. https://deanattali.com/shinyjs/

BSBI. (2017a). *Ellenberg indicator r (reaction)*. https://database.bsbi.org/object.php?entityid=2cd4p9h.2cg1m1&class=ChecklistInstance

BSBI. (2017b). *Ellenberg indicator s (salt tolerance)*. https://database.bsbi.org/object.php?entityid=2cd4p9h.2cg1s5&class=ChecklistInstance

BSBI. (2017c). *Ellenberg indicator n (nitrogen)*. https://database.bsbi.org/object.php?entityid=2cd4p9h.2cg1yq&class=ChecklistInstance

BSBI. (2017d). *Ellenberg indicator f (moisture)*. https://database.bsbi.org/object.php?entityid=2cd4p9h.2cg1kd&class=ChecklistInstance

BSBI. (2017e). *Ellenberg indicator l (light)*. https://database.bsbi.org/object.php?entityid=2cd4p9h.2cg1v7&class=ChecklistInstance

Bunce, R. G. H., Barr, C. J., Gillespie, M. K., Howard, D. C., Scott, W. A., Smart, S. M., Van de Poll, H. M., & Watkins, J. W. (1999). *Vegetation of the British countryside - the Countryside Vegetation System. ECOFACT Volume 1* (Vol. 1). DETR. ISBN: 978-1-85112-155-7

Butcher, B., Carey, P., Edmonds, R., Norton, L., & Treweek, J. (2020). *UK habitats classification user manual version 1.1*. Stockport: UKHab Ltd.

Chang, W., Cheng, J., Allaire, J., Sievert, C., Schloerke, B., Xie, Y., Allen, J., McPherson, J., Dipert, A., & Borges, B. (2024). *Shiny: Web application framework for r*. https://shiny.posit.co/

Cheng, J., Sievert, C., Schloerke, B., Chang, W., Xie, Y., Allen, J., Software, P., & PBC. (2024). *Htmltools: Tools for HTML*.

DRAFT

CIEEM. (2022). *Guidelines for ecological impact assessment in the UK and ireland. Terrestrial, freshwater, coastal and marine. September 2018. Version 1.2 - updates april 2022*. PDF. https://cieem.net/wp-content/uploads/2018/08/ECIA-Guidelines-2018-Terrestrial-Freshwater-Coastal-and-Marine-V1.2-April-22-Compressed.pdf

Gann, G. D., McDonald, T., Walder, B., Aronson, J., Nelson, C. R., Jonson, J., Hallett, J. G., Eisenberg, C., Guariguata, M. R., Liu, J., Hua, F., Echeverría, C., Gonzales, E., Shaw, N., Decleer, K., & Dixon, K. W. (2019). International principles and standards for the practice of ecological restoration. Second edition. *Restoration Ecology*, *27*(S1), S1–S46. https://doi.org/10.1111/rec.13035

Hill, M. O. (1989). Computerized matching of relevés and association tables, with an application to the British National Vegetation Classification. *Vegetatio*, *83*(1), 187–194. https://doi.org/10.1007/BF00031691

Hill, M. O., Preston, C. D., Bosanquet, S. D. S., & B., R. D. (2007). *BRYOATT - attributes of british and irish mosses, liverworts and hornworts - spreadsheet*. Centre for Ecology; Hydrology. https://www.brc.ac.uk/biblio/bryoatt-attributes-british-and-irish-mosses-liverworts-and-hornwo

JNCC. (2008). *Spreadsheet of habitat correspondences*. https://hub.jncc.gov.uk/assets/9e70531b-5467-4136-88f6-3b3dd905b56d

Kass, J. M., Pinilla-Buitrago, G. E., Paz, A., Johnson, B. A., Grisales-Betancur, V., Meenan, S. I., Attali, D., Broennimann, O., Galante, P. J., Maitner, B. S., Owens, H. L., Varela, S., Aiello-Lammens, M. E., Merow, C., Blair, M. E., & Anderson, R. P. (2023). Wallace 2: A shiny app for modeling species niches and distributions redesigned to facilitate expansion via module contributions. *Ecography*, *2023*(3), e06547. https://doi.org/10.1111/ecog.06547

Lin, G. (2024). *Reactable: Interactive Data Tables for R*. https://glin.github.io/reactable/

Maciejewski, L., Pinto, P. E., Wurpillot, S., Drapier, J., Cadet, S., Muller, S., Agou, P., Renaux, B., & Gégout, J.-C. (2020). Vegetation unit assignments: Phytosociology experts and classification programs show similar performance but low convergence. *Applied Vegetation Science*, *23*(4), 698–709. https://doi.org/10.1111/avsc.12516

Malloch, A. J. C. (1998). *MATCH version 2*. University of Lancaster.

Marrs, R. H. (2019). *TABLEFIT v. 3.0 & v.4, programs for the identification of vegetation types according to the British National Vegetation Classification Includes software* [Other]. Centre for Ecology & Hydrology. https://www.ceh.ac.uk/sites/default/files/Tablefit-Aug2016.zip

Meyer, F., & Perrier, V. (2024). *Shinybusy: Busy Indicators and Notifications for 'Shiny' Applications*. https://github.com/dreamRs/shinybusy

Mucina, L., Bültmann, H., Dierßen, K., Theurillat, J.-P., Raus, T., Čarni, A., Šumberová, K., Willner, W., Dengler, J., García, R. G., Chytrý, M., Hájek, M., Di Pietro, R., Iakushenko, D., Pallas, J., Daniëls, F. J. A., Bergmeier, E., Santos Guerra, A., Ermakov, N., … Tichý, L. (2016). Vegetation of Europe: Hierarchical floristic classification system of vascular plant, bryophyte, lichen, and algal communities. *Applied Vegetation Science*, *19*(S1), 3–264. https://doi.org/10.1111/avsc.12257

NBN Trust. (2024a). *Anthocerotophyta child taxa*. https://species.nbnatlas.org/species/NHMSYS0021386659#classification

NBN Trust. (2024b). *Bryophyta child taxa*. https://species.nbnatlas.org/species/NBNSYS0000172125#classification

NBN Trust. (2024c). *Marchantiophyta child taxa*. https://species.nbnatlas.org/species/NHMSYS0020796639#classification

NBN Trust. (2024d). *Pteridophyta child taxa*. https://species.nbnatlas.org/species/

DRAFT

NHMSYS0001747256#classification

NBN Trust. (2024e). *Tracheophyta child taxa*. https://species.nbnatlas.org/species/NHMSYS0020787695#classification

Oksanen, J., Simpson, G., Solymos, P., Lahti, L., Hannigan, G., Weedon, J., Szöcs, E., McGlinn, D., Legendre, P., Zhilong, Bolker, B., Chirico, M., Friendly, M., Stier, A., Borcard, D., Dunné, F. van, Bengtsson, H., Andersen, K. S., Barbour, M., … brendanf. (2022). *Vegan: Community Ecology Package*. Zenodo. https://doi.org/10.5281/zenodo.7185692

Ooms, J. (2024). *Writexl: Export Data Frames to Excel 'xlsx' Format*. https://ropensci.r-universe.dev/writexl https://docs.ropensci.org/writexl/

Owen, J. (2021). *Rhandsontable*. https://github.com/jrowen/rhandsontable

Perrier, V., Meyer, F., & Granjon, D. (2024). *shinyWidgets: Custom Inputs Widgets for Shiny*. https://github.com/dreamRs/shinyWidgets

Perrin, P. (2019). *ERICA - Engine for Relevés to Irish Communities Assignment V5.0 User's Manual*. BEC Consultants Ltd. https://biodiversityireland.shinyapps.io/vegetation-classification/_w_6fddd58f/manual.pdf

Perrin, P., Fitzpatrick, Ú., & Lynn, D. (2018). The Irish Vegetation Classification - an overview of concepts, structure and tools. *CIEEM in Practice*, *102*, 14–19. http://www.botanicalenvironmental.com/wp-content/uploads/2018/12/IVC-CIEEM-Article.pdf

Pescott, O., Walker, K., Harris, F., New, H., Cheffings, C., Newton, N., Jitlal, M., Redhead, J., Smart, S. M., & Roy, D. B. (2019). *S1 file* [Data set]. https://www.researchgate.net/publication/333011121_S1_File

Prosser, M., Wallace, H., & Gowing, D. (2023). Phytosociology informs the conservation of species-rich meadows in hydrologically dynamic habitats: an example from British floodplains in a wider European context. *British & Irish Botany*, *5*(1). https://doi.org/10.33928/bib.2023.05.001

Pywell, R. F., Bullock, J. M., Hopkins, A., Walker, K. J., Sparks, T. H., Burke, M. J. W., & Peel, S. (2002). Restoration of species-rich grassland on arable land: Assessing the limiting processes using a multi-site experiment. *Journal of Applied Ecology*, *39*(2), 294–309. https://doi.org/10.1046/j.1365-2664.2002.00718.x

R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Raper, C. (2015). *UK Species Inventory - Master copy*. https://doi.org/10.5519/0038741

Ridding, L. E., Bullock, J. M., Pescott, O. L., Hawes, P., Walls, R., Pereira, M. G., Thacker, S. A., Keenan, P. O., Dragosits, U., & Pywell, R. F. (2020). Long-term change in calcareous grassland vegetation and drivers over three time periods between 1970 and 2016. *Plant Ecology*, *221*(5), 377–394. https://doi.org/10.1007/s11258-020-01016-1

Rodwell, J. S. (Ed.). (1991). *British Plant Communities: Volume 1: Woodlands and Scrub* (Vol. 1). Cambridge University Press. https://doi.org/10.1017/9780521235587

Rodwell, J. S. (Ed.). (1992a). *British Plant Communities: Volume 2: Mires and Heaths* (Vol. 2). Cambridge University Press. https://doi.org/10.1017/9780521391658

Rodwell, J. S. (Ed.). (1992b). *British Plant Communities: Volume 3: Grasslands and Montane Communities* (Vol. 3). Cambridge University Press. https://doi.org/10.1017/9780521391665

Rodwell, J. S. (Ed.). (1995). *British Plant Communities: Volume 4: Aquatic Communities, Swamps and Tall-Herb Fens* (Vol. 4). Cambridge University Press. https://doi.org/10.1017/9781107340817

DRAFT

Rodwell, J. S. (Ed.). (2000). *British Plant Communities: Volume 5: Maritime Communities and Vegetation of Open Habitats* (Vol. 5). Cambridge University Press. https://doi.org/10.1017/CBO9780511541834

Rodwell, J. S. (Ed.). (2006). *National vegetation classification: Users' handbook*. Joint Nature Conservation Committee (JNCC) Peterborough. https://data.jncc.gov.uk/data/a407ebfc-2859-49cf-9710-1bde9c8e28c7/JNCC-NVC-UsersHandbook-2006.pdf

Sievert, C. (2020). *Interactive Web-Based Data Visualization with R, plotly, and shiny*. Chapman and Hall/CRC. https://doi.org/10.1201/9780429447273

Sievert, C., Cheng, J., & Aden-Buie, G. (2024). *Bslib: Custom 'Bootstrap' 'Sass' Themes for 'shiny' and 'rmarkdown'*. https://rstudio.github.io/bslib/

Sievert, C., PBC, P. S., & Otto, M. (2023). *Bsicons: Easily Work with 'Bootstrap' Icons*.

Smart, S. M. (2000). Ecological assessment of vegetation from a nature reserve using regional reference data and indicator scores. *Biodiversity and Conservation*, *9*(6), 811–832. https://doi.org/10.1023/a:1008948602316

Smart, S. M., Goodwin, A., & Wallace, H. (2016). *MAVIS (ver 1.03) user manual* (S. M. Smart, Ed.). https://www.ceh.ac.uk/sites/default/files/Mavis-Desktop-Help.pdf

Smart, S. M., Wood, C. M., Seaton, F. M., McMullen, A., Ward, S. E., Ward, R., O'Reilly, J., Ruffino, L., Losse, P., Stroh, P., Pilkington, S., Saunders, L., Wallace, H. L., O'Rourke, C., Long, C. E., Potter, R., Maskell, L. C., Sier, A. R. J., Norton, L. R., … Vallis-Wilks, M. (2024). *Broadleaved woodland survey of great britain, 2020-2022*. NERC EDS Environmental Information Data Centre. https://doi.org/10.5285/42C203C8-44DE-40E2-A694-B1E8CBD4C8E1

Stefan, S. M., Wickham, H., Henry, L., & RStudio. (2022). *Magrittr: A Forward-Pipe Operator for R*.

Sturbois, A., De Cáceres, M., Bifolchi, A., Bioret, F., Boyé, A., Gauthier, O., Grall, J., Grémare, A., Labrune, C., Robert, A., Schaal, G., & Desroy, N. (2023). Ecological quality assessment: A framework to report ecosystems quality and their dynamics from reference conditions. *Ecosphere*, *14*(12), e4726. https://doi.org/10.1002/ecs2.4726

Vaidyanathan, R., Xie, Y., Allaire, J. J., Cheng, J., Sievert, C., Russell, K., Hughes, E., Software, P., & PBC. (2023). *Htmlwidgets: HTML Widgets for R*.

Wallace, H., Gowing, D., Lawson, C., Rothero, E., & Strachan, I. (2023). An exploration of oceanic wet grasslands in the Scottish coastal lowlands. *British Wildlife*. https://www.britishwildlife.com/article/article-volume-35-number-3-page-193-200/

Wallace, H., & Jones, L. (2018). *Monitoring of forest removal and groundworks to rejuvenate dunes at newborough warren - ynys llanddwyn SSSI. Vegetation change, third year results for 2017*. NERC/Centre for Ecology & Hydrology. http://nora.nerc.ac.uk/id/eprint/524094/

Wallace, H., & Prosser, M. (2017). *A review of the national vegetation classification for the calthion group of plant communities in england and wales (JP021)*. Natural England. https://publications.naturalengland.org.uk/publication/5839929072943104

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., … Yutani, H. (2019). Welcome to the Tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Wood, C. M., Smart, S. M., & Bunce, R. G. H. (2015). Woodland survey of great britain 1971-2001. *Earth System Science Data*, *7*(2), 203–214. https://doi.org/10.5194/essd-7-203-2015

DRAFT

Xie, Y. (2023). *Markdown: Render Markdown with 'commonmark'*. https://cran.r-project.org/web/packages/markdown/index.html

Xie, Y. (2024). *Bookdown: Authoring Books and Technical Documents with R Markdown*. https://github.com/rstudio/bookdown

Zhu, H. (2024). *kableExtra*. https://github.com/haozhu233/kableExtra

DRAFT